Biliary Atresia, Part Six: Adjuvant Therapy after the Kasai Operation. Steroids, Immunoglobulin, Antibiotics and Ursodeoxycholic Acid.

Imagine a girl who is forty-five days old. Two days ago she had a Kasai portoenterostomy for biliary atresia. The operation went well: the surgeon found a type three atresia, transected the fibrous plate flush with the liver capsule and brought up a Roux limb. Now she is back on the ward, and the questions begin. The junior doctor asks whether she should start steroids. A parent has read online about immunoglobulin infusions. The surgical team has written for an oral antibiotic to prevent infection, and the pharmacy has already dispensed ursodeoxycholic acid. Which of these medicines will change whether this child keeps her own liver? This chapter is about that question.

The main idea can be stated simply. No adjuvant therapy, meaning no medicine given after the operation to support it, has strong proof that it improves survival with the native liver. Steroids lower the bilirubin faster, mainly when the Kasai is done early, but they cost the infant early serious complications and slower growth. Intravenous immunoglobulin and the two drugs that block the ileal bile acid transporter have failed in trials. Antibiotic prophylaxis and ursodeoxycholic acid are used almost everywhere, but they rest on small and weak studies. The outcome is still decided mostly by the operation itself, by the age at which it is done and by the experience of the centre.

A few terms first. The Kasai portoenterostomy is the operation that removes the scarred bile duct remnant at the porta hepatis, the hilum of the liver, and drains the tiny surviving bile channels into a loop of small bowel called the Roux limb. Clearance of jaundice means that the bilirubin falls to normal or near normal after the operation. Different studies use slightly different thresholds, such as below 1.5 milligrams per decilitre or below 2 milligrams per decilitre. Native-liver survival means being alive with one's own liver, without a transplant. Cholangitis is a bacterial infection of the bile ducts inside the liver; after a Kasai it is common and it can stop bile flow for good. A serious adverse event is any harmful event in a trial that is life-threatening, needs hospital care or prolongs it.

The sources for this chapter include the Steroids in Biliary Atresia Randomized Trial, known as START, published by Bezerra and colleagues in the Journal of the American Medical Association in 2014; the growth follow-up of that trial by Alonso and colleagues in 2018; the British trial of low-dose prednisolone by Davenport and colleagues in 2007 and the later high-dose study from King's College Hospital in 2013; the Cochrane review of steroids from 2018; a larger meta-analysis by Yang and colleagues in 2022; an open-label randomised trial from Shanghai by Lu and colleagues in 2023; the immunoglobulin trial known as PRIME, published by Mack and colleagues in 2019; studies of antibiotic prophylaxis from Taiwan, China and Israel; an international expert consensus on cholangitis from 2022; a withdrawal

study of ursodeoxycholic acid from Lille in 2008; and company announcements about odevixibat and maralixibat.

Part one. What an adjuvant would have to fix.

To judge any medicine given after the Kasai, it helps to picture what continues to go wrong in the liver once the operation is over. The disease does not stop when the remnant is removed. Think of it as a chain of five links.

The first link is ongoing inflammation. The portal tracts of an infant with biliary atresia are full of activated immune cells, mainly T lymphocytes and macrophages, and this inflammation continues for weeks after surgery. Steroids and immunoglobulin are aimed here. The second link is injury to the surviving ductules at the porta. These microscopic channels are the only route for bile to leave the liver, and if they swell, scar or close, drainage stops. Steroids are thought to reduce the swelling and scarring around them, and newer agents such as N-acetylcysteine and granulocyte colony-stimulating factor aim to protect or repair them. The third link is poor bile flow, with retained bile acids and jaundice. Ursodeoxycholic acid, and the drugs that reduce the bile acid pool by blocking reabsorption in the ileum, are aimed here. The fourth link is ascending infection. Because the Roux limb connects the porta directly to the bowel, bacteria can travel up into the liver. Antibiotic prophylaxis is aimed at this link. The fifth link is fibrosis, the scar that accumulates with each insult and leads to cirrhosis and portal hypertension. No medicine currently targets it directly.

The important point is that these links are in series, one after another. A drug that calms inflammation cannot open ductules that the surgeon did not reach. A drug that improves bile flow cannot help if no bile is draining at all. This explains two things that we will see again and again. First, the strongest predictors of outcome remain the age at the Kasai, the anatomy found at the porta and the experience of the centre. Second, the effect of any single adjuvant is likely to be modest, so trials need many patients to detect it.

Part two. Steroids: four regimens, one question.

Steroids have been given after the Kasai for decades. The reasoning is that they are anti-inflammatory and may limit scarring, and that they may reduce swelling around the ductules in the first weeks after surgery. Over twenty years the doses used in studies have risen, because the early low-dose trial showed only a faster early fall in bilirubin.

Four regimens are worth knowing. The first British randomised trial, published by Davenport and colleagues in 2007, used oral prednisolone at 2 milligrams per kilogram per day, tapered over about six weeks, in 71 infants in two centres. The American START trial used a higher dose: intravenous methylprednisolone at 4 milligrams per kilogram per day for the first three days, then oral prednisolone at 4 milligrams per kilogram per day until day fourteen, then 2 milligrams per kilogram per day for two weeks, then 1 milligram per kilogram per day for two weeks, and then a weekly taper through 0.8, 0.6, 0.4, 0.2 and 0.1 milligrams per kilogram per

day, finishing with alternate-day doses and stopping at about week fourteen. The King's College Hospital high-dose protocol, reported in 2013, started oral prednisolone at 5 milligrams per kilogram per day, giving a total of about 75 milligrams per kilogram over the course, roughly twice the total of the low-dose regimen. And an open-label trial from Shanghai, published in 2023, gave steroids for ten to twelve weeks to infants with type three atresia. When you read pooled analyses, remember that the dose, the day steroids are started and the age at the Kasai all differ from study to study.

Part three. The START trial.

START is the largest and most rigorous trial of steroids after the Kasai. It was double-blind and placebo-controlled, meaning that neither the families nor the doctors knew who received the drug. It enrolled 140 infants, 70 in each group, between 2005 and 2011 in the North American Childhood Liver Disease Research Network. Infants were randomised within 72 hours of the operation. Their mean age was 2.3 months, which is late compared with many European programmes. The primary endpoint, the main result the trial was designed to measure, was a bilirubin below 1.5 milligrams per decilitre with the native liver at six months.

The results can be described as three pairs of bars. For the primary endpoint, 58.6 percent of the steroid group reached a bilirubin below 1.5 at six months, compared with 48.6 percent of the placebo group. That is a gain of about ten percentage points, and after adjustment the difference was 8.7 percent, with a wide confidence interval from minus 10.4 percent to plus 27.7 percent, and a p value of 0.43. In plain words, the result was compatible with no effect, with a moderate benefit, or even with slight harm. For survival with the native liver at 24 months, the two groups were essentially identical: 58.7 percent with steroids and 59.4 percent with placebo, with a hazard ratio of 1.0. And for safety, the first serious adverse event occurred within 30 days of surgery in 37.2 percent of the steroid group, compared with 19.0 percent of the placebo group, a difference that was statistically significant. Across the whole study the total rate of serious adverse events was similar, about 81 percent against 80 percent, which tells us that the extra harm was concentrated in the weeks of treatment.

Surgical complications deserve a specific mention. Six infants in the steroid group had a serious surgical adverse event, including problems at the anastomosis and intestinal perforation, compared with one infant in the placebo group. This is biologically plausible: high-dose steroids slow wound healing and can mask the signs of a leak.

Many people quote a subgroup result from START. Among the 76 infants who had their Kasai at 70 days of age or younger, 71.8 percent of the steroid group cleared their jaundice, compared with 56.8 percent of the placebo group. This looks like a large benefit, but it is a post hoc analysis, meaning that it was not planned in advance, the p value was 0.36, and the statistical test for whether age changed the effect of steroids was far from significant, with a p value of 0.67. It is a hypothesis, not a finding.

The price of steroids became clearer in the growth follow-up published by Alonso and colleagues in 2018. Infants who received steroids had lower length and head circumference measurements, expressed as z-scores, for the first three months after the Kasai. Their weight remained lower until twelve months of age. Head growth caught up later in the first year. Strikingly, the growth penalty was largest in infants whose Kasai worked well, which is exactly the group in whom steroids are most often recommended. Infants with biliary atresia are already malnourished, because they cannot absorb fat and fat-soluble vitamins, and poor growth predicts worse outcomes at transplant. A drug that slows linear growth for three months has to buy something real. START also reported that an inadequate response to immunisations occurred in 51.5 percent of steroid-treated infants against 38.5 percent of placebo infants; this difference was not significant, but live vaccines should not be given while an infant is on high-dose steroids.

Part four. The European signal and the Shanghai trial.

The British experience tells a slightly different story. The 2007 trial of low-dose prednisolone found no difference in clearance or native-liver survival, but there was a faster fall in bilirubin at one month in infants operated before 70 days. That observation led the King's College team to try a higher dose. In their 2013 study, 44 infants operated at 70 days or younger received high-dose prednisolone and were compared with infants who received no steroids. Clearance of jaundice was 66 percent with high-dose steroids against 52 percent without them, a difference with a p value of 0.037. Native-liver survival was 56 percent against 48 percent, which was not significantly different. A later analysis by Tyraskis and Davenport in 2016, covering 104 infants on high-dose steroids, found a strong age gradient: all eleven infants operated before 30 days cleared their jaundice, compared with about two thirds of those operated at 61 to 70 days, and the age effect was seen mainly in isolated biliary atresia. These are important observations, but they are not randomised, they come from a single surgeon in a single centre, and they used historical as well as contemporary controls.

The most positive result comes from Shanghai. Lu and colleagues randomised 200 infants with type three atresia to steroids or no steroids after the Kasai, 100 in each group. At six months, 54.1 percent of the steroid group were jaundice-free with their native liver, against 31.0 percent of the control group. At 24 months, native-liver survival was 57.1 percent against 40.0 percent. Serious adverse events within three months were not significantly different. This is the only randomised trial to show a difference in native-liver survival. But it has two weaknesses that matter. It was open-label, so the clinicians and assessors knew which infants received steroids. And the control group did unusually poorly: only 31 percent cleared their jaundice at six months, compared with 48.6 percent in the placebo arm of START. Part of the gap may therefore reflect a weak comparison group rather than a strong drug. The trial strengthens the hypothesis that steroids help some infants, but it has not been replicated in a blinded multicentre trial.

Part five. What the pooled evidence says.

Two meta-analyses reach different conclusions, and the reason is instructive. The Cochrane review by Tyraskis, Parsons and Davenport in 2018 included only the two placebo-controlled randomised trials, the British trial and START, with about 211 infants in total. For death or transplant at two years, the risk ratio was 1.00, with a confidence interval from 0.72 to 1.39, which means no effect. For failure to clear jaundice at six months, the risk ratio was 0.89, with a confidence interval from 0.67 to 1.17, which is a small and non-significant trend in favour of steroids. The authors calculated that the trials had collected only about 39 percent of the information needed to answer the clearance question reliably. They rated all the evidence as low certainty and concluded that larger, multicentre, placebo-controlled trials are needed.

The meta-analysis by Yang and colleagues in 2022 cast a wider net: five randomised trials and six cohort studies, with 1,032 infants. It found that steroids increased clearance at six months, with a risk ratio of 1.35, meaning about a third more infants cleared, and increased native-liver survival at 24 months, with a risk ratio of 1.31, although not at six or twelve months. The effect on cholangitis was not clear. The benefit was concentrated in infants operated before 70 days and in medium-dose regimens starting intravenously.

So the answer depends on what you allow in. If you include only blinded placebo-controlled trials, there is no effect on survival and an underpowered, non-significant gain in clearance. If you add open-label trials and cohort studies, which carry a higher risk of bias, the result becomes positive. Both analyses agree on two points: any benefit is modest, and it is most plausible in infants operated early.

Part six. Using steroids sensibly.

Given this evidence, steroids are an option and not a standard of care. If a centre chooses to use them, the trials suggest a decision frame. Factors that lean towards steroids include a Kasai before 70 days and ideally before 45 days, isolated atresia, good-sized ductules at the porta, an uncomplicated operation with bile appearing early in the stool, no sepsis, good feeding and normal growth, and a centre with a protocol that monitors growth and adverse events. Factors that lean against steroids include a Kasai after 90 days with established cirrhosis, the splenic malformation syndrome, a complicated operation with bowel injury, leak or reoperation, early cholangitis or line sepsis, and growth that is already faltering. If steroids are used, they should start once the anastomosis is secure, growth and infection should be watched closely, and live vaccines should be deferred while the dose is high.

Part seven. Intravenous immunoglobulin.

The rationale for intravenous immunoglobulin came from the immune hypothesis of biliary atresia. In mouse models in which atresia is induced by rhesus rotavirus, immunoglobulin reduced injury to the bile duct cells and the inflammation around them. The Childhood Liver Disease Research Network tested this in a trial called PRIME, published by Mack and

colleagues in 2019. It was an open-label phase one and two trial, meaning that it was designed mainly to test safety and feasibility, and it had no control group of its own. Infants received immunoglobulin at 1 gram per kilogram on days three to five, 30 and 60 after the Kasai, and 29 infants completed the study. The comparison group was the placebo arm of START.

The treatment was feasible, with an acceptability of 79 percent, and serious adverse events occurred in 90 percent of infants, none of which were attributed to immunoglobulin, which reflects how sick these infants are. But there was no increase in the proportion with a bilirubin below 1.5 milligrams per decilitre at 90, 180 or 360 days. Survival with the native liver at one year was 58.6 percent with immunoglobulin, compared with 70.5 percent in the historical placebo group, a difference of almost 12 percentage points in the wrong direction. The programme did not go on to a larger trial. The lesson is that a single-arm trial compared with historical controls can only rule out a large effect, but a large effect is what this trial was looking for, and it did not find one.

Part eight. Antibiotic prophylaxis.

Cholangitis is the most common complication after the Kasai, affecting roughly 30 to 60 percent of children, most often in the first two years. Each episode can damage the ductules and end bile drainage. So the idea of preventing it with a daily oral antibiotic is attractive. The evidence, however, is thin.

The study most often quoted is from Taiwan, published by Bu and colleagues in 2003. It included 19 infants who had already had one episode of cholangitis, so it tested secondary rather than primary prevention. They were randomised to trimethoprim-sulfamethoxazole or to neomycin and compared with 18 historical controls. Both antibiotics reduced recurrences compared with the controls, with p values of 0.042 and 0.011, and the two drugs did not differ from each other. There has never been a randomised placebo-controlled trial of oral prophylaxis after the Kasai.

Other studies are of mixed quality and mixed results. A randomised trial from China in 2021, with 180 infants, compared seven against fourteen days of intravenous antibiotics after surgery. Early cholangitis was less common with the longer course, 38 percent against 61 percent, but the rate at six months was similar, and the authors concluded that the short course was sufficient. A study of 72 infants from Israel in 2025 found that primary prophylaxis, given to about half of them, did not lower the rate of infection. The rationale for prophylaxis is strong, but its effect size is unknown, and the counterweights are antibiotic resistance and Clostridioides difficile infection.

In the absence of trials, experts have tried to agree on practice. An international Delphi consensus on cholangitis after the Kasai, published by Calinescu-Tuleasca and colleagues in 2022 within the European rare liver disease network, supported oral prophylaxis for six to twelve months after the operation. There was no agreement on which drug to use, although

trimethoprim-sulfamethoxazole was the most common choice, used by about 54 percent of respondents. The experts agreed that suspected cholangitis should be treated for ten to fourteen days and confirmed cholangitis, meaning proven by culture or imaging, for fourteen to twenty-one days. In Switzerland, a national protocol introduced in 2015 was followed by fewer recurrences and fewer transplants. Consensus is the weakest form of evidence, but standardising definitions, cultures, the empirical intravenous regimen and the duration lets a centre audit itself. A prospective international study called TRACK-BA began collecting data in 2025 to validate the definitions that future trials of prophylaxis will need.

Part nine. Ursodeoxycholic acid.

Ursodeoxycholic acid, usually shortened to ursodiol, is a hydrophilic bile acid that makes bile less toxic to cells and stimulates bile flow. It is given to almost every infant after the Kasai, but it has never been tested in a randomised trial in biliary atresia.

The only direct test is a withdrawal study from Lille, published by Willot and colleagues in 2008. Sixteen children who had a successful Kasai and had taken ursodiol for at least 18 months, at a mean dose of 25 milligrams per kilogram per day, stopped it, and then restarted it. Liver tests worsened in 13 of the 16. The alanine aminotransferase rose from about one and a half times the upper limit of normal to three times. The gamma-glutamyl transferase rose from about four times to eight times the upper limit. The aspartate aminotransferase rose from about 1.3 to 1.7 times. All improved after ursodiol was resumed. So in children who already drain bile, ursodiol keeps the liver tests lower. Whether that translates into better long-term native-liver survival is unknown. Typical doses are 10 to 20 milligrams per kilogram per day. It is not a substitute for drainage, and very high doses deserve caution, because in adults with primary sclerosing cholangitis a dose of 28 to 30 milligrams per kilogram per day was harmful.

Part ten. Practice variation.

Because the evidence is weak, practice varies widely. In a survey of 41 centres in China, 90 percent used steroids, all used antibiotics and 93 percent used ursodiol, often for more than a year. In Canada, 24 percent of paediatric gastroenterologists used steroids, while 95 percent used ursodiol and 80 percent prescribed oral antibiotics at discharge. In an analysis of 504 infants in 37 American children's hospitals between 2017 and 2024, about 28 percent received steroids. Steroids were given more often at high-volume hospitals, they lengthened the hospital stay, and two-year native-liver survival did not differ. In short, steroid use follows local trial results and local culture more than evidence, while ursodiol and prophylaxis are near-universal. This variation also means that comparisons of outcomes between countries are confounded by different adjuvant mixes.

Part eleven. What to expect next.

The newest hope was the class of drugs that block the ileal bile acid transporter, the protein in the terminal ileum that reabsorbs bile acids. Blocking it reduces the bile acid pool and relieves

itch in Alagille syndrome and progressive familial intrahepatic cholestasis, conditions in which the bile ducts still drain. Two programmes have now tested this idea after the Kasai. Maralixibat was tested in a phase two trial called EMBARK, in infants who had their Kasai before 90 days of age, for 26 weeks; in December 2023 the company announced that it had missed its primary endpoint, the change in bilirubin. Odevixibat was tested in a large phase three trial called BOLD, with 254 infants in 19 countries, for up to two years, with transplant or death as the primary endpoint; in July 2026 the company announced that it had not met this endpoint. In biliary atresia the problem is the outflow itself, so it is perhaps not surprising that reducing the bile acid pool did not change the course. These drugs should stay inside trials after the Kasai.

Other agents are still being studied. Intravenous N-acetylcysteine, given for seven days after the Kasai, is being tested in a small phase two study at Baylor College of Medicine, with normalisation of serum bile acids at 24 weeks as the main outcome. Granulocyte colony-stimulating factor, given as three daily doses, is being tested in a phase two randomised trial in Vietnam, Pakistan and the United States. Results of both are pending. The next generation of adjuvants is aimed at the survival and repair of ductules rather than at the bile acid pool.

Part twelve. What the guidelines say.

No major society has a guideline on medical care after the Kasai. The American and European adult liver societies, the AASLD and the EASL, have no biliary atresia treatment guideline. The 2017 guideline on the cholestatic infant from the North American and European paediatric gastroenterology societies, known as NASPGHAN and ESPGHAN, covers diagnosis and does not address adjuvants. The Japanese Biliary Atresia Society guideline from 2021 states that a recommendation on steroids could not be made because of a scarcity of evidence. The only structured guidance on antibiotics comes from the 2022 Delphi consensus. So the most-used medicines after the Kasai are prescribed without any formal recommendation. The documents are aligned in not supporting immunoglobulin or ileal bile acid transporter inhibitors outside trials, they leave gaps on prophylaxis and cholangitis treatment that only consensus fills, and they differ most on steroids, where practice ranges from near-universal use to almost none.

Part thirteen. Traps and pearls.

Several errors in reasoning are common. The first is believing that steroids can rescue a late Kasai. Any benefit is confined to early operations; in an older infant with cirrhosis, only the harm remains. The second is starting high-dose steroids on the first day without regard to the operation; early serious adverse events doubled in START, and surgical complications were six times as frequent. The third is attributing an individual child's clearance to steroids: about half of infants clear without them, so in any single child it is impossible to know. The fourth is assuming that a child on prophylaxis cannot have cholangitis. Fever after a Kasai is cholangitis until proven otherwise and needs blood cultures and intravenous antibiotics. The fifth is adding

an ileal bile acid transporter inhibitor in the hope of improving drainage; two trials in atresia were negative. And the last, perhaps the most important, is treating a rising bilirubin at two or three months with more medicines instead of referral. Failure to clear jaundice by three months predicts the need for transplant, and no adjuvant changes that trajectory. The right adjuvant for a failing Kasai is a transplant assessment.

Let us return to a few clinical scenarios. A 38-day-old with isolated atresia and good bile flow on day five is the best-case group, and steroids are a reasonable option after a frank discussion with the family. A 95-day-old with the splenic malformation syndrome and a firm liver should not receive steroids, and transplant planning should start early. An infant on day two after the Kasai with ileus and a bile-stained drain should not start steroids. Parents who ask about immunoglobulin can be told that it was tested and showed no benefit. An infant on prophylactic trimethoprim-sulfamethoxazole who develops fever at six weeks needs cultures and ten to fourteen days of intravenous antibiotics. A child who has cleared jaundice and whose family wants to stop ursodiol can stop it, with liver tests repeated after four to eight weeks, and restart it if the tests rise. And a child whose bilirubin is still 5 milligrams per decilitre at three months, despite every adjuvant, needs a transplant assessment now.

So, tomorrow in clinic, carry these points. In START, steroids raised clearance at six months from 48.6 to 58.6 percent, a difference that was not significant, and did not change native-liver survival at two years, about 59 percent in both groups. Steroids doubled serious adverse events in the first 30 days, from 19 to 37 percent, and slowed length and head growth for three months and weight for a year. Any steroid benefit is most plausible in infants operated before about 70 days, and the positive Shanghai trial was open-label with a weak control group. The Cochrane review found low-certainty evidence and no effect on death or transplant. Immunoglobulin showed no benefit, with one-year native-liver survival of about 59 percent against 71 percent in historical controls. Odevixibat and maralixibat both failed after the Kasai. Prophylaxis for six to twelve months is expert consensus rather than proven practice, and cholangitis is treated for ten to fourteen days when suspected and fourteen to twenty-one days when confirmed. Ursodiol keeps liver tests lower in children who drain bile, but has no outcome data. Above all, the medicine that matters most after a Kasai is still a well-timed operation in an experienced centre, and early referral when it fails.